

Solution Code - Module 3-Core Java - HTTPClientAPI

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q36_HTTPClientAPI.java

```
1 | import java.net.URI;
2 | import java.net.http.HttpClient;
3 | import java.net.http.HttpRequest;
4 | import java.net.http.HttpResponse;
5 | import java.time.Duration;
6 |
7 | public class Q36_HTTPClientAPI {
8 |     public static void main(String[] args) {
9 |         System.out.println("=== HTTP Client API (Java 11+) ===");
10 |
11 |         // Using a public, stable API endpoint
12 |         String apiUri = "https://httpbin.org/get";
13 |         System.out.println("Sending GET request to: " + apiUri);
14 |
15 |         // 1. Build the HttpClient
16 |         HttpClient client = HttpClient.newBuilder()
17 |             .connectTimeout(Duration.ofSeconds(10))
18 |             .build();
19 |
20 |         // 2. Build the HttpRequest
21 |         HttpRequest request = HttpRequest.newBuilder()
22 |             .uri(URI.create(apiUri))
23 |             .header("Accept", "application/json")
24 |             .header("User-Agent", "Java HttpClient Demo")
25 |             .GET()
26 |             .build();
27 |
28 |         // 3. Send request and print response
29 |         try {
30 |             HttpResponse<String> response = client.send(request, HttpResponse.BodyHandlers.ofString());
31 |
32 |             System.out.println("\n--- Response Status Code ---");
33 |             System.out.println("Status: " + response.statusCode());
34 |
35 |             System.out.println("\n--- Response Headers ---");
36 |             response.headers().map().forEach((k, v) -> System.out.println(k + ": " + v));
37 |
38 |             System.out.println("\n--- Response Body ---");
39 |             System.out.println(response.body());
40 |
41 |         } catch (Exception e) {
42 |             System.err.println("HTTP Request failed: " + e.getMessage());
43 |             e.printStackTrace();
44 |         }
```

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```
44 |         }  
45 |     }  
46 | }
```